

Homework 1

Victoria Lam Linear Algebra Sep 18, 2026

Problem 1

(a)

Final answer: 24

(b)

$$(1/2) w = [-2, 1, 3]^T$$

$$v + (1/2) w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T$$

$$\| [-1, 5, 1]^T \| = \sqrt{1 + 25 + 1} = \sqrt{27} = 3\sqrt{3} \approx 5.20$$

$$[\text{ANSWER: } 3\sqrt{3} \approx 5.20]$$

(c)

$$\cos \theta = 2/7 \text{ so } \theta \approx 73.40^\circ$$

Problem 2

(a)

AB is undefined: A is 3×3 and B is 2×3 , so A has 3 columns but B has 2 rows, $3 \neq 2$.

BA works: B is 2×3 and A is 3×3 , so BA is 2×3 .

row 1: $[(1)(2) + (0)(1) + (-2)(0), (1)(-1) + (0)(3) + (-2)(4), (1)(0) + (0)(-2) + (-2)(1)]$

row 2: $[(3)(2) + (1)(1) + (1)(0), (3)(-1) + (1)(3) + (1)(4), (3)(0) + (1)(-2) + (1)(1)]$

[ANSWER: $BA = \begin{bmatrix} 2 & -9 & -2 \\ 7 & 4 & -1 \end{bmatrix}$]

(b)

AV : A is 3×3 and V is 3×1 , so AV is 3×1 .

$AV = [(2)(1) + (-1)(-2) + (0)(3), (1)(1) + (3)(-2) + (-2)(3), (0)(1) + (4)(-2) + (1)(3)]^T$

$= [4, -11, -5]^T$

VA : multiplication is commutative for this, so $VA = AV^T = [4 \ -11 \ -5]$.

[ANSWER: $AV = [4, -11, -5]^T$, $VA = [4 \ -11 \ -5]$]

(c)

Aw is undefined. A has 3 columns and w has only 1 row, and $3 \neq 1$.

WA : w is 1×3 and A is 3×3 , so WA is 1×3 .

$WA = [(2)(2) + (0)(1) + (-1)(0), (2)(-1) + (0)(3) + (-1)(4), (2)(0) + (0)(-2) + (-1)(1)]$

[ANSWER: Aw undefined, $WA = [4 \ -6 \ -1]$]

(d)

vw : v is 3×1 and w is 1×3 , so vw is 3×3 (outer product).

$$vw = \begin{bmatrix} (1)(2) & (1)(0) & (1)(-1) \\ (-2)(2) & (-2)(0) & (-2)(-1) \\ (3)(2) & (3)(0) & (3)(-1) \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ -4 & 0 & 2 \\ 6 & 0 & -3 \end{bmatrix}$$

wv : w is 1×3 times v 3×1 , so wv is 1×1 : $wv = (2)(1) + (0)(-2) + (-1)(3) = -1$

[ANSWER: vw as above, $wv = [-1]$]

(e)

$A^2 = AA$, and A is square so this is fine.

$$\text{row 1: } [(2)(2) + (-1)(1) + (0)(0), (2)(-1) + (-1)(3) + (0)(4), (2)(0) + (-1)(-2) + (0)(1)]$$

[

$$\text{row 2: } [(1)(2) + (3)(1) + (-2)(0), (1)(-1) + (3)(3) + (-2)(4), (1)(0) + (3)(-2) + (-2)(1)]$$

[

$$\text{row 3: } [(0)(2) + (4)(1) + (1)(0), (0)(-1) + (4)(3) + (1)(4), (0)(0) + (4)(-2) + (1)(1)]$$

]

[ANSWER: $A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}$]

(f)

B^2 is undefined: B is 2×3 , so BB needs the 3 columns of the first B to match the 2 rows of the second, and $3 \neq 2$. Only square matrices can be squared.

Problem 3

(a)

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_1: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R_2 \rightarrow -R_2: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_2: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 10 & | & 10 \end{bmatrix}$$

$$R_3 \rightarrow (1/10)R_3: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & -3 & | & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & | & 1 \end{bmatrix}$$

Back substitute: $x_3 = 1$, $x_2 = -4 + 3(1) = -1$, $x_1 = -1 - 2(-1) + 1 = 2$. One solution.

$$\text{[ANSWER: } x = [2, -1, 1]^T \text{]}$$

(b)

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 & 1 & | & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \\ 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \\ 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \\ 3 & 7 & 0 & | & 10 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1: \begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \\ 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \\ 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \\ 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2: \begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2: \begin{bmatrix} 1 & 0 & -7 & | & 8 \\ 0 & 1 & 3 & | & -2 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & | & -2 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$$

No pivot in column 3, so z is free and there are infinitely many solutions. Let $z = t$.

Row 1 gives $x - 7t = 8$ and row 2 gives $y + 3t = -2$, so $x = 8 + 7t$, $y = -2 - 3t$.

[ANSWER: $[x, y, z]^T = [8, -2, 0]^T + t[7, -3, 1]^T, t \in \mathbb{R}$]

Problem 4

(a)

Let x_1, x_2, x_3 = kilograms (kg) of Alloy 1, Alloy 2, and Alloy 3 used.

The 100 kg batch is 36% copper, 44% zinc, 20% nickel, so it holds 36 kg copper, 44 kg zinc, 20 kg nickel. Nothing is lost, so metal in = metal out:

$$\text{copper: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 36$$

$$\text{zinc: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 44$$

$$\text{nickel: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 20$$

(b)

Scaling every equation by 10 makes the entries whole numbers:

$$[5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

(c)

$$[5 \ 2 \ 4 \mid 360] \quad [1 \ 0 \ -4 \mid -40] \quad [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440] \rightarrow [3 \ 7 \ 2 \mid 440] \rightarrow [0 \ 7 \ 14 \mid 560]$$

$$[2 \ 1 \ 4 \mid 200] \quad [2 \ 1 \ 4 \mid 200] \quad [2 \ 1 \ 4 \mid 200]$$

$$[1 \ 0 \ -4 \mid -40] \quad [1 \ 0 \ -4 \mid -40] \quad [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 7 \ 14 \mid 560] \rightarrow [0 \ 1 \ 2 \mid 80] \rightarrow [0 \ 1 \ 2 \mid 80]$$

$$[0 \ 1 \ 12 \mid 280] \quad [0 \ 1 \ 12 \mid 280] \quad [0 \ 0 \ 10 \mid 200]$$

$$[1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 80]$$

$$[0 \ 0 \ 1 \mid 20]$$

So $x_3 = 20$, then $x_2 = 80 - 2(20) = 40$, then $x_1 = -40 + 4(20) = 40$.

check: $40 + 40 + 20 = 100$ kg total, and copper is $20 + 8 + 8 = 36$ kg.

[ANSWER: 40 kg of Alloy 1, 40 kg of Alloy 2, 20 kg of Alloy 3]

Problem 5

(a)

$$\begin{array}{ccccc} [1 \ 2 \ 1] & [1 \ 2 \ 1] & [1 \ 2 \ 1] & [1 \ 2 \ 1] & [1 \ 2 \ 0] \\ [2 \ 4 \ 3] \rightarrow [0 \ 0 \ 1] & \rightarrow [0 \ 0 \ 1] & \rightarrow [0 \ 0 \ 1] & \rightarrow [0 \ 0 \ 1] & \rightarrow [0 \ 0 \ 1] \\ [3 \ 6 \ 4] & [3 \ 6 \ 4] & [0 \ 0 \ 1] & [0 \ 0 \ 0] & [0 \ 0 \ 0] \end{array}$$

[ANSWER: $\text{RREF}(A) = [1 \ 2 \ 0; 0 \ 0 \ 1; 0 \ 0 \ 0]$]

(b)

$\text{rank}(A) = 2$, because the RREF has exactly two pivots, in column 1 and column 3 (equivalently, two nonzero rows).

(c)

0 solutions: possible. Running the same operations on $[A \mid b]$ turns the bottom row into $[0 \ 0 \ 0 \mid b_3 - b_1 - b_2]$. If $b_3 \neq b_1 + b_2$, say $b = [0, 0, 1]^T$, that row says $0 = 1$, which is impossible, so the system is inconsistent.

Exactly 1 solution: impossible. Column 2 is never a pivot column, so x_2 is a free variable whenever the system is consistent, and $\text{rank}(A) = 2 < 3$ unknowns.

Infinitely many: possible. If $b_3 = b_1 + b_2$, say $b = [0, 0, 0]^T$ or $b = [1, 3, 4]^T$, the system is consistent and the free variable x_2 gives a whole line of solutions.

Problem 6

(a)

These are dependent because there are three vectors and v_3 has a zero entry, so they can't fill $\mathbb{R}^3$.

[ANSWER: dependent; span is a plane]

(b)

Put v_1, v_2, v_3 in as columns and row reduce:

$$\begin{array}{l} [1 \ 0 \ 1] \quad [1 \ 0 \ 1] \quad [1 \ 0 \ 1] \\ [1 \ 1 \ 0] \rightarrow [0 \ 1 \ -1] \rightarrow [0 \ 1 \ -1] \\ [0 \ 1 \ 1] \quad [0 \ 1 \ 1] \quad [0 \ 0 \ 2] \\ (R_2 \rightarrow R_2 - R_1, \text{ then } R_3 \rightarrow R_3 - R_2) \end{array}$$

Every column has a pivot, so $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ forces $c_1 = c_2 = c_3 = 0$ and the

set is linearly independent. Three independent vectors in $\mathbb{R}^3$ span the whole space.

[ANSWER: independent; span is all of $\mathbb{R}^3$]